

Exercise 1: PDB Practical: Exploring Protein Structure

Objective: To develop familiarity with the Protein Data Bank (PDB) and use structural information to answer biological/biophysical questions.

Instructions: Use the **RCSB Protein Data Bank (PDB)** to answer the following questions. Record the relevant PDB ID, observations, and numerical values where applicable.

1. Finding a structure

Search the PDB for **human hemoglobin**.

- Find the **PDB ID** of one experimentally determined structure.
- What experimental method was used to determine the structure?
- What is its reported resolution?

For the PDB ID selected above: Answer the following

2. How many protein chains are present ? How many chains make up the biologically functional hemoglobin molecule?
3. Select one globin chain. What is its chain ID? How many amino-acid residues does it contain? What is the first and last residue ?
4. Which type of secondary structure predominates: **α -helices, β -sheets, or turns/loops**? Identify one α -helix and suggest how long it is and the residues it contains.
5. **Identify any ligand present.** What is its PDB three-letter ligand code?
6. Which amino-acid residue is directly coordinated to the central metal ion? Estimate the distance between the **iron atom and the coordinating residue**. What is the type of interaction?
7. Compare two globin chains within the same hemoglobin structure. Are their sequences identical? Identify **one residue position that differs**, if applicable. Does the structural fold remain broadly similar or differ ?
8. Choose one buried residue and one surface-exposed residue in the globin chain. What structural features suggest that each residue is buried or exposed? Which residue would you expect to have greater interaction with solvent? What type of amino acids would you generally expect to find in each environment?
9. Locate the **heme-binding pocket** in the structure. Identify two amino-acid residues surrounding the heme.
10. Find another PDB structure of human hemoglobin determined using a different experimental method or at a different resolution. Give its PDB ID. Are there any major structural difference you notice? Which structure would you consider more suitable for **precise analysis of side-chain positions**, and why?